HW 4

Kevin Lin

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(a) By the cone condition (0.1) $\|\Delta_{S^c}\|_1 \leq 3\|\Delta_S\|_1$. Since S and S^c partition the coordinates of Δ, we have:

$$\begin{aligned}\|\Delta\|_1 &= \|\Delta_S\|_1 + \|\Delta_{S^c}\|_1 \\ &\leq \|\Delta_S\|_1 + 3\|\Delta_S\|_1 \\ &= 4\|\Delta_S\|_1\end{aligned}$$

Because $|S| = k$, applying the Cauchy-Schwarz inequality gives:

$$\begin{aligned}\|\Delta_S\|_1 &\leq \sum_{i\in S} |\Delta_i| \\ &\leq \sqrt{k}\sqrt{\sum_{i\in S} |\Delta_i|^2} \\ &= \sqrt{k}\|\Delta_S\|_2\end{aligned}$$

Combining the two inequalities gives us $\|\Delta\|_1 \leq 4\sqrt{k}\|\Delta_S\|_2$. Substituting the ℓ_2 bound (0.2) gives us:

$$\begin{aligned}\|\Delta\|_1 &\leq 4\sqrt{k}\left(\frac{3\lambda\sqrt{k}}{\kappa}\right) \\ &= \frac{12\lambda k}{\kappa}\end{aligned}$$

Thus, $\|\Delta\|_1 \leq \frac{12\lambda k}{\kappa}$.

(b) For any $q \in [1, 2]$, we can define the Holder conjugates $a = \frac{1}{2-q}$ and $b = \frac{1}{q-1}$ such that $\frac{1}{a} + \frac{1}{b} = 1$. By the definition of the ℓ_q norm, we have $\|v\|_q^q = \sum_{i=1}^p |v_i|^q$. We can rewrite each term $|v_i|^q = |v_i|^{2-q}|v_i|^{2(q-1)}$ and

apply Holder's inequality to get:

$$\begin{aligned}
\sum_{i=1}^{p} |v_i|^q &= \sum_{i=1}^{p} |v_i|^{2-q} |v_i|^{2(q-1)} \\
&\leq \left(\sum_{i=1}^{p} |v_i|^{(2-q)a} \right)^{\frac{1}{a}} \left(\sum_{i=1}^{p} |v_i|^{2(q-1)b} \right)^{\frac{1}{b}} \\
\|v\|_q^q &= \left(\sum_{i=1}^{p} |v_i| \right)^{2-q} \left(\sum_{i=1}^{p} |v_i|^2 \right)^{q-1} \\
&= \|v\|_1^{2-q} \|v\|_2^{2(q-1)} \\
\|v\|_q &\leq \|v\|_1^{\frac{2-q}{q}} \|v\|_2^{\frac{2(q-1)}{q}} \\
&= \|v\|_1^{2/q-1} \|v\|_2^{2-2/q}
\end{aligned}$$

For $q = 1, 2$, the inequality holds with equality, thus $\|v\|_q \leq \|v\|_1^{2/q-1} \|v\|_2^{2-2/q}$ for all $q \in [1, 2]$. Using the bounds from (a) and (0.2), $\|\Delta\|_1 \leq \frac{12\lambda k}{\kappa}$ and $\|\Delta\|_2 \leq \frac{3\lambda\sqrt{k}}{\kappa}$. We can substitue these bounds into the inequality to get:

$$\begin{aligned}
\|\Delta\|_q &\leq \left(\frac{12\lambda k}{\kappa} \right)^{2/q-1} \left(\frac{3\lambda\sqrt{k}}{\kappa} \right)^{2-2/q} \\
&= \left(\frac{\lambda}{\kappa} \right)^{\frac{2}{q}-1+2-\frac{2}{q}} \cdot k^{\frac{2}{q}-1+\frac{2-2/q}{2}} \cdot 12^{\frac{2}{q}-1} \cdot 3^{2-\frac{2}{q}} \\
&= \left(\frac{\lambda}{\kappa} \right) \cdot k^{1/q} \cdot 4^{2/q-1} \cdot 3 \\
&= 3 \cdot 4^{2/q-1} \frac{k^{1/q}\lambda}{\kappa}
\end{aligned}$$

For $q = 1$, the constant is $3 \cdot 4^{2-1} = 12$, and for $q = 2$, the constant is $3 \cdot 4^{1-1} = 3$, so both are always ≤ 12. Thus, we have $\|\Delta\|_q \leq \frac{12k^{1/q}\lambda}{\kappa}$ for all $q \in [1, 2]$. When $q = 1$, we have $\|\Delta\|_1 \leq \frac{12k\lambda}{\kappa}$ which recovers (a), and when $q = 2$, we have $\|\Delta\|_2 \leq \frac{3\sqrt{k}\lambda}{\kappa}$ which recovers (0.2).

(c) Utilize (0.1) $\frac{1}{2n}\|X\Delta\|_2^2 \leq \frac{3\lambda}{2}\|\Delta_S\|_1 - \frac{\lambda}{2}\|\Delta_{S^c}\|_1$. Because $\|Delta_{S^c}\|_1 \geq 0$, we can drop the second term on the right-hand side to get:

$$\begin{aligned}
\frac{1}{2n}\|X\Delta\|_2^2 &\leq \frac{3\lambda}{2}\|\Delta_S\|_1 \\
\frac{1}{n}\|X\Delta\|_2^2 &\leq 3\lambda\|\Delta_S\|_1
\end{aligned}$$

By Cauchy-Schwarz, since $|S| = k$, $\|Delta_S\|_1 \leq \sqrt{k}\|\Delta_S\|_2 \leq \sqrt{k}\|\Delta\|_2$. Substituting this into the inequality gives $\frac{1}{n}\|X\Delta\|_2^2 \leq 3\lambda\sqrt{k}\|\Delta\|_2$. We

can then apply the ℓ_2 bound (0.2) to get:

$$\begin{aligned}\frac{1}{n}\|X\Delta\|_2^2 &\leq 3\lambda\sqrt{k}\left(\frac{3\lambda\sqrt{k}}{\kappa}\right)\\ &= \frac{9k\lambda^2}{\kappa}\end{aligned}$$

Thus, $\frac{1}{n}\|X\Delta\|_2^2 \leq \frac{9k\lambda^2}{\kappa}$.

(d) With the assumptions, by the optimality of the LASSO estimator, the objective value is no larger than the objective value β^*:

$$\frac{1}{2n}\|y - X\hat{\beta}\|_2^2 + \lambda\|\hat{\beta}\|_1 \leq \frac{1}{2n}\|y - X\beta^*\|_2^2 + \lambda\|\beta^*\|_1$$

We know that $y = X\beta^* + \epsilon$, $\hat{\beta} = \beta^* + \Delta$, thus $y - X\hat{\beta} = \epsilon - X\Delta$ and $y - X\beta^* = \epsilon$. Substituting these into the inequality gives:

$$\begin{aligned}\frac{1}{2n}\|\epsilon - X\Delta\|_2^2 + \lambda\|\beta^* + \Delta\|_1 &\leq \frac{1}{2n}\|\epsilon\|_2^2 + \lambda\|\beta^*\|_1\\ \|\epsilon - X\Delta\|_2^2 &= \|\epsilon\|_2^2 - 2\epsilon^\top X\Delta + \|X\Delta\|_2^2\\ \frac{1}{2n}\|X\Delta\|_2^2 + \lambda\|\hat{\beta}\|_1 &\leq \frac{1}{n}\epsilon^\top X\Delta + \lambda\|\beta^*\|_1\end{aligned}$$

Using Holder's inequality and the noise condition:

$$\begin{aligned}\frac{1}{n}\epsilon^T X\Delta &\leq \left(\frac{1}{n}X^\top\epsilon\right)^\top\Delta\\ &\leq \|\frac{1}{n}X^\top\epsilon\|_\infty\|\Delta\|_1\\ &\leq \frac{\lambda}{2}\|\Delta\|_1\end{aligned}$$

Thus, $\frac{1}{2n}\|X\Delta\|_2^2 + \lambda\|\hat{\beta}\|_1 \leq \frac{\lambda}{2}\|\Delta\|_1 + \lambda\|\beta^*\|_1$. By the triangle inequality, $\|\Delta\|_1 = \|\hat{\beta} - \beta^*\|_1 \leq \|\hat{\beta}\|_1 + \|\beta^*\|_1$, so we can substitute this into the inequality to get:

$$\begin{aligned}\frac{1}{2n}\|X\Delta\|_2^2 + \lambda\|\hat{\beta}\|_1 &\leq \frac{\lambda}{2}(\|\hat{\beta}\|_1 + \|\beta^*\|_1) + \lambda\|\beta^*\|_1\\ &= \frac{\lambda}{2}\|\hat{\beta}\|_1 + \frac{3\lambda}{2}\|\beta^*\|_1\\ \frac{1}{2n}\|X\Delta\|_2^2 + \frac{\lambda}{2}\|\hat{\beta}\|_1 &\leq \frac{3\lambda}{2}\|\beta^*\|_1\\ \frac{1}{n}\|X\Delta\|_2^2 + \lambda\|\hat{\beta}\|_1 &\leq 3\lambda\|\beta^*\|_1\end{aligned}$$

Because $\lambda\|\hat{\beta}\|_1 \geq 0$, we can drop this term to get $\frac{1}{n}\|X\Delta\|_2^2 \leq 3\lambda\|\beta^*\|_1$. Because the prediction error is nonnegative, $\frac{1}{n}\|X\Delta\|_2^2 \geq 0$, we can drop

this term to get $\lambda\|\hat{\beta}\|_1 \leq 3\lambda\|\beta^*\|_1$. We know $\lambda > 0$, so we can get $\|\hat{\beta}\|_1 \leq 3\|\beta^*\|_1$.

Now, taking $\lambda \asymp \sigma\sqrt{\frac{\log p}{n}}$ and assuming β^* has k nonzero entries, each equal to ± 1, $\|\beta^*\|_1 = \sum i = 1^p|\beta_i^*| = k$. Then the fast rate from (c):

$$\begin{aligned}\frac{1}{n}\|X\Delta\|_2^2 &\leq \frac{9k\lambda^2}{\kappa} \\ &\lesssim \frac{9k\sigma^2\log p}{\kappa n}\end{aligned}$$

For fixed k, p, σ, κ, this is $\mathcal{O}(1/n)$.
The slow rate from (d):

$$\begin{aligned}\frac{1}{n}\|X\Delta\|_2^2 &\leq 3\lambda\|\beta^*\|_1 = 3k\lambda \\ &\lesssim 3k\sigma\sqrt{\frac{\log p}{n}}\end{aligned}$$

For fixed k, p, σ, this is $\mathcal{O}(1/\sqrt{n})$.
Comparing the bounds:

$$\begin{aligned}\frac{9k\lambda^2}{\kappa} &\leq 3k\lambda \\ \frac{3\lambda}{\kappa} &\leq 1 \\ \lambda &\leq \frac{\kappa}{3}\end{aligned}$$

We know $\lambda \asymp \sigma\sqrt{\frac{\log p}{n}}$, so $\sigma\sqrt{\frac{\log p}{n}} \leq \frac{\kappa}{3}$. Rearrange for n gives $n \geq \frac{9\sigma^2\log p}{\kappa^2}$. Thus, the fast rate is better than the slow rate when $n \gtrsim \frac{9\sigma^2\log p}{\kappa^2}$.

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(a) Let $u \in \mathbb{R}^p$ such that $\|u\|_2 = 1$, $\|u\|_0 \leq k$. Let $T = \text{supp}(u)$, then $|T| \leq k$. Then, by the triangle inequality and definitions of $\delta_1(A)$ and

$\mu(A)$:

$$\begin{aligned}
\|Au\|_2^2 - 1 &= u^\top (A^\top A - I)u \\
&= \sum_{j \in T} u_j^2 (\|A_j\|_2^2 - 1) + \sum_{i,j \in T, i \neq j} u_i u_j A_i^\top A_j \\
&\leq \delta_1(A) \sum_{j \in T} u_j^2 + \mu(A) \sum_{i,j \in T, i \neq j} |u_i||u_j| \\
&= \delta_1(A) + \mu(A) \left[\left(\sum_{j \in T} |u_j| \right)^2 - \sum_{j \in T} u_j^2 \right] \\
&= \delta_1(A) + \mu(A) \left[\|u\|_1^2 - 1 \right]
\end{aligned}$$

Since u has at most k nonzero coordinates, by Cauchy-Schwarz, $\|u\|_1 \leq \sqrt{k}\|u\|_2 = \sqrt{k}$. Therefore, $\|u\|_1^2 - 1 \leq k - 1$, and we have $|\|Au\|_2^2 - 1| \leq \delta_1(A) + (k-1)\mu(A)$. Taking the supremum over every k-sparse unit vector u gives us $\delta_k(A) \leq \delta_1(A) + (k-1)\mu(A)$.

(b) We start by bounding $\delta_1(A)$. For every column j, $\|A_j\|_2^2 - 1 = \frac{1}{n}\sum_{i=1}^n (X_{ij}^2 - 1)$. Since $X_{ij} \sim \mathcal{N}(0,1)$, $E[X_{ij}^2 - 1] = 0$, and the centered squares are sub-exponential. If we apply Bernstein's inequality, there then exists an absolute constant $c > 0$ such that:

$$P(|\|A_j\|_2^2 - 1| \geq t) \leq 2\exp(-cnt^2) \quad \forall 0 < t \leq 1$$

Applying the union bound over all p columns gives:

$$\begin{aligned}
P(\delta_1(A) \geq t) &= P\left(\max_{j=1,\ldots,p} |\|A_j\|_2^2 - 1| \geq t \right) \\
&\leq \sum_{j=1}^p P(|\|A_j\|_2^2 - 1| \geq t) \\
&\leq 2p\exp(-cnt^2)
\end{aligned}$$

Now we bound $\mu(A)$. For distinct columns $i \neq j$, $A_i^\top A_j = \frac{1}{n}\sum_{k=1}^n X_{ki}X_{kj}$. Because each entry is independent and nonzero, $E[X_{ki}X_{kj}] = 0$, and the product of independent Gaussians is sub-exponential. Thus, again with Bernstein's inequality, we have $P(|A_i^\top A_j| \geq t) \leq 2\exp(-cnt^2)$ for some absolute constant $c > 0$ and $0 < t \leq 1$. There are $\binom{p}{2}$ distinct pairs of

columns, thus applying the union bound gives:

$$\begin{aligned} P(\mu(A) \geq t) &= P\left(\max_{i,j=1,\ldots,p,i\neq j} |A_i^\top A_j| \geq t\right) \\ &\leq \sum_{i,j=1,\ldots,p,i\neq j} P(|A_i^\top A_j| \geq t) \\ &\leq 2\binom{p}{2}\exp(-cnt^2) \\ &\leq p^2\exp(-cnt^2) \end{aligned}$$

Now we combine the two bounds:

$$\begin{aligned} P(\delta_1(A) \geq t \text{ or } \mu(A) \geq t) &\leq 2p\exp(-cnt^2) + p^2\exp(-cnt^2) \\ &\leq 2p^2\exp(-cnt^2), \quad p \geq 2 \end{aligned}$$

Choose $t = C\sqrt{\frac{\log p}{n}}$ with $C = \sqrt{3/c}$. Then $cnt^2 = 3\log p$, and we have:

$$\begin{aligned} P(\delta_1(A) \geq C\sqrt{\frac{\log p}{n}} \text{ or } \mu(A) \geq C\sqrt{\frac{\log p}{n}}) &\leq 2p^2\exp(-3\log p) \\ &= 2p^2p^{-3} = 2/p \end{aligned}$$

For $n \geq C'\log p$, with $C' \geq C^2$, we have $t \leq 1$, and thus with probability at least $1 - 2/p$, $\delta_1(A) \leq C\sqrt{\frac{\log p}{n}}$ and $\mu(A) \leq C\sqrt{\frac{\log p}{n}}$.

(c) From (a), we have $\delta_k(A) \leq \delta_1(A) + (k-1)\mu(A)$. From (b), with probability at least $1 - 2/p$, $\delta_1(A) \leq C\sqrt{\frac{\log p}{n}}$ and $\mu(A) \leq C\sqrt{\frac{\log p}{n}}$. Thus, with probability at least $1 - 2/p$:

$$\begin{aligned} \delta_k(A) &\leq C\sqrt{\frac{\log p}{n}} + (k-1)C\sqrt{\frac{\log p}{n}} \\ &= kC\sqrt{\frac{\log p}{n}} \end{aligned}$$

To guarantee k-RIP with constant δ, we require $kC\sqrt{\frac{\log p}{n}} \leq \delta$, which rearranges to $n \geq C^2\delta^{-2}k^2\log p$. Taking $C'' = C^2$, or a larger absolute constant as necessary, gives $n \geq C''\delta^{-2}k^2\log p$. Thus, for $0 < \delta < 1$, this sample size guarantees k-RIP with probability at least $1 - 2/p$.
k is lost in (a) when we bound all off diagonal entires using the worst case coherence: $\sum_{i\neq j} |u_i||u_j|| \langle A_i, A_j\rangle | \leq (k-1)\mu(A)$. Since $\mu(A) \lesssim \sqrt{\frac{\log p}{n}}$, we need to multiply by k to account for the k nonzero entries in u. This is a loose bound because taking the absolute value ignores cancellation among the off diagonal entries, and thus we get $\delta_k(A) \lesssim$

$k\sqrt{\frac{\log p}{n}}$. Squaring the resulting sample size condition introduces the k^2 term.

For a fixed k-sparse unit vector u, Bernstein's inequality gives:

$$P(|\|Au\|_2^2 - 1| \geq \delta) \leq 2\exp(-cn\delta^2)$$

for $0 < \delta < 1$. A 1/4-net for the k-spare unit sphere has size at most $|\mathcal{N}| \leq 9^k\binom{p}{k} \leq \left(\frac{9ep}{k}\right)^k$. Applying the union bound and the net-to-sphere argument after adjusting absolute constants gives $P(\delta_k(A) \geq \delta) \leq 2\exp\left(-cn\delta^2 + k\log(9ep/k)\right)$. Thus, k-RIP holds with high probability when $n \gtrsim \delta^{-2}k\log(ep/k)$. Unlike the coherence argument, the net argument controls all sparse directions instead of summing over the worst case coherence, and thus avoids the loose k factor.